

SwishNet: Predicting NBA Shot Outcomes with Spatio-Temporal GNNs

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Predicting

NBA shot outcomes are stochastic, but the spatial context surrounding each shot carries predictive signal that tabular features miss. We present **SwishNet**, predicting make/miss from SportVU tracking data on **87,147 shots**. The input is a spatio-temporal graph $\mathcal{G}$ built from (x, y, z) positions at shot release and 2–4 preceding snapshots; the output is $\hat{p} \in [0, 1]$. We benchmark LR, decision trees, and XGBoost against four GNN operators and 29 HP configs. Our hybrid pipeline (GNN + GMM + XGBoost) achieves **AUC 0.6551**.

Data, Features & Graph Construction

2015–16 NBA SportVU: (x, y, z) of 10 players + ball at 25 Hz, 632 games, aligned with play-by-play and 21 per-player stats. **87,147 shots** (FG% 45.8%). GNN: 61K/17K/8.7K; baselines: 5-fold CV.

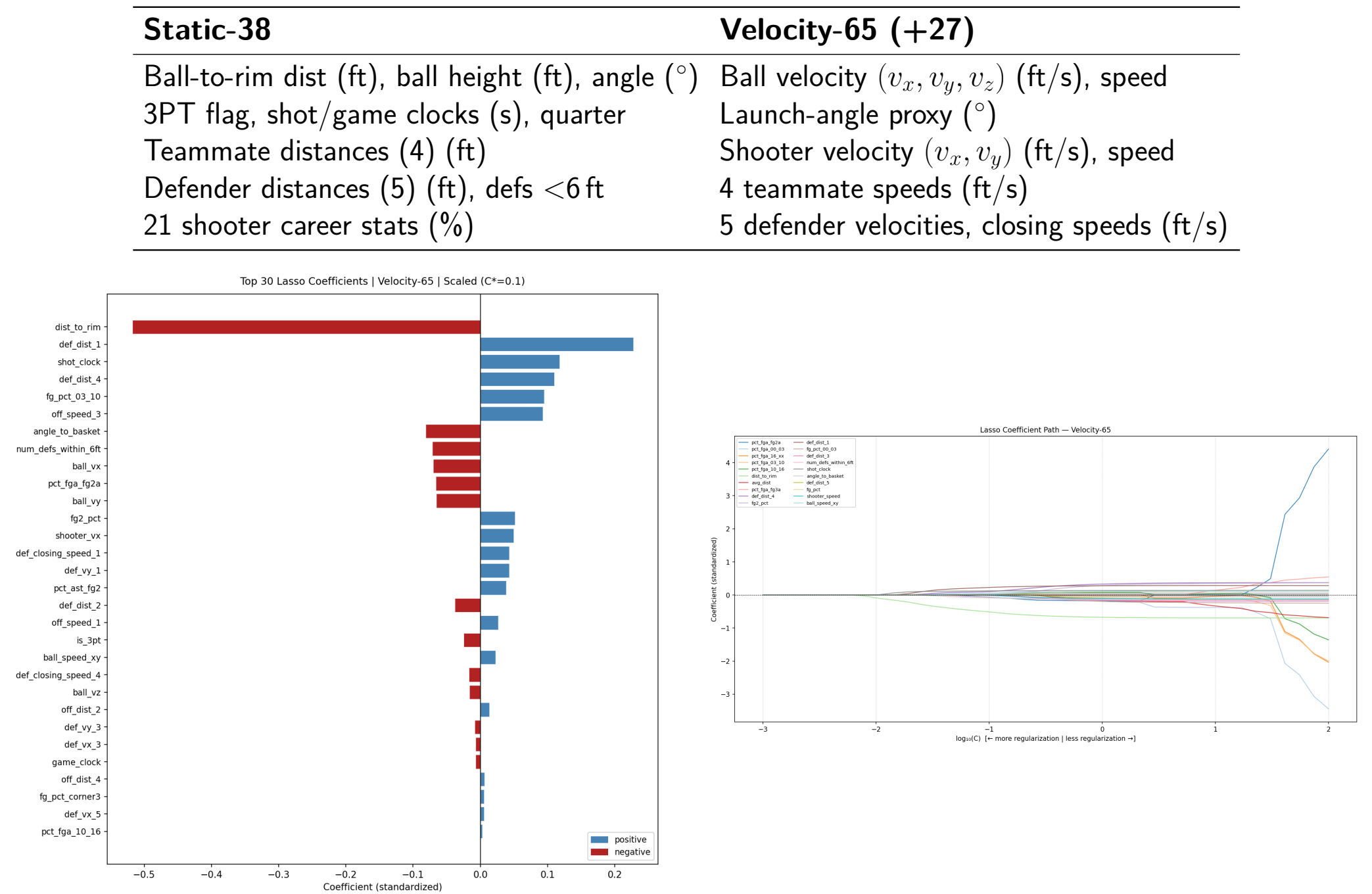


Figure 1. Left: Top-30 Lasso coefficients. Right: Lasso path vs. regularization.

Graph

A physics-based algorithm scans backward from PBP events to locate the release frame via rim-moment detection and shooter identification. Per timestep: 11 fully-connected nodes (5 off + 5 def + ball). Default: 3 snapshots at offsets [3, 12] frames behind release (0.12 s and 0.48 s at 25 Hz), yielding 33 nodes and ~ 330 edges.

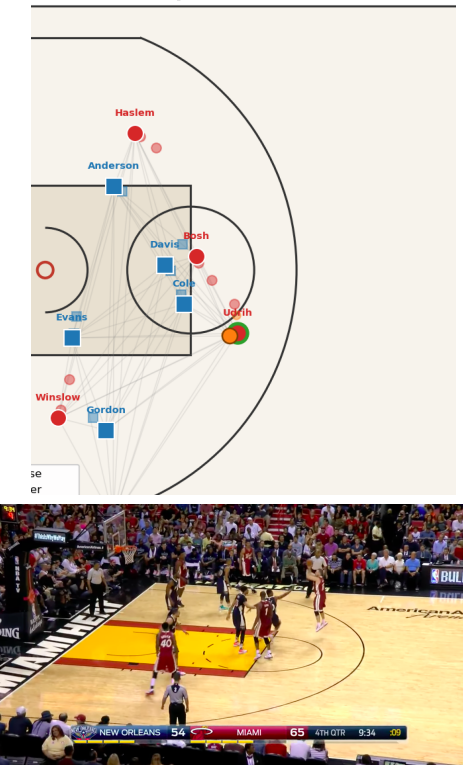


Figure 2. Udrih mid-range miss, MIA vs NOP: graph (top) and game footage (bottom).

Table 1. Node and edge feature vectors for each spatio-temporal graph.

Node Features (41 per node)		
x, y, z	Court position (ft)	
has_ball, is_offense	Binary role flags	
is_player_node	1 for players, 0 for ball	
dist_to_rim, angle_to_basket	Geometry (ft, $^\circ$)	
dist_to_ball_handler	Spacing (ft)	
dist_to_3pt_line	Court zone (ft)	
num_nearby_defenders	Count within 6 ft	
quarter, game_clock, shot_clock	Game state (s)	
timestamp	Temporal index (0–2)	
position_encoding (5)	Learnable node ID	
player_stats (21)	Career shooting %	
Edge Features (9 per edge)		
$\Delta x, \Delta y$	Relative position (ft)	
euclidean_dist	Pairwise distance (ft)	
edge_angle	Bearing ($^\circ$)	
type: OO, OD, DD, PB	One-hot relation type	
type: TEMPORAL	Temporal link flag	

Models

Logistic Regression

$p(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$ where $\sigma(z) = 1/(1 + e^{-z})$. Fit by minimizing cross-entropy with L_1 (Lasso) or L_2 (Ridge) regularization; strength $C = 1/\lambda$ selected via inner 3-fold CV.

XGBoost [1]

Additive ensemble $\hat{y}_i = \sum_{t=1}^T f_t(\mathbf{x}_i)$, trained by gradient boosting. $T=400$ trees, depth=4, $\eta=0.05$.

GATv2Conv [2]

Dynamic attention applied *after* concatenation:

$$\alpha_{ij} = \frac{\exp(\mathbf{a}^\top \text{LeakyReLU}(\mathbf{W}[\mathbf{h}_i \parallel \mathbf{h}_j \parallel \mathbf{e}_{ij}]))}{\sum_{k \in \mathcal{N}(i)} \exp(\mathbf{a}^\top \text{LeakyReLU}(\mathbf{W}[\mathbf{h}_i \parallel \mathbf{h}_k \parallel \mathbf{e}_{ik}]))}$$

GINEConv

Injective neighborhood aggregation with edge features:

$$\mathbf{h}'_i = \text{MLP}\left((1+\epsilon) \mathbf{h}_i + \sum_{j \in \mathcal{N}(i)} \text{ReLU}(\mathbf{h}_j + \mathbf{e}_{ij})\right)$$

Temporal GRU Aggregation [3]

Per-timestep GATv2/GINE $\rightarrow$ mean pool $\rightarrow \mathbf{h}_t \in \mathbb{R}^d$. GRU aggregates the sequence: $\mathbf{z}_t = \sigma(\mathbf{W}_z[\mathbf{h}_t \parallel \mathbf{s}_{t-1}])$, $\mathbf{s}_t = (1 - \mathbf{z}_t) \odot \mathbf{s}_{t-1} + \mathbf{z}_t \odot \tanh(\mathbf{W}[\mathbf{h}_t \parallel \mathbf{r}_t \odot \mathbf{s}_{t-1}])$

Final $\mathbf{s}_T \in \mathbb{R}^{256}$ classified by 3-layer MLP.

PCA + Gaussian Mixture Model

We reduce 256-dim GRU embeddings via PCA (88.4% variance in 2 PCs), then fit a GMM ($K=10$, BIC-selected):

$$p(\mathbf{z}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

Each shot receives a 10-dim soft membership vector $[\gamma_{i,1}, \dots, \gamma_{i,10}]$.

Hybrid Pipeline: GNN + GMM + XGBoost

The GMM recovers 10 NBA shot archetypes *without supervision* from the GNN's latent space. We concatenate the 10-dim soft membership vector with Static-38 features and retrain XGBoost, yielding **AUC 0.6551**—our best result.

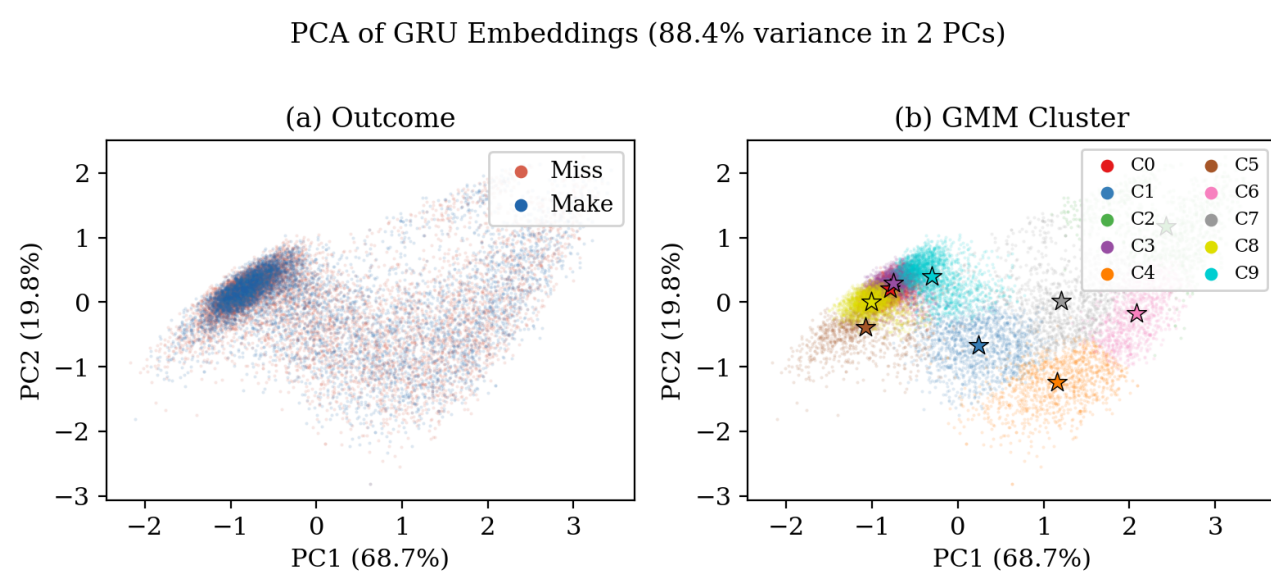


Figure 3. (a) PCA by outcome; (b) by GMM cluster.

Cl.	n	FG%	Dist	%3PT	Clock	Archetype
2	5,461	87.7	5.7	0.0	18.5	At-rim / dunks
6	5,415	65.2	2.7	0.0	15.9	Putbacks
7	6,049	62.0	5.8	0.7	13.9	Short-range
1	7,755	41.1	7.1	0.1	11.5	Paint
0	11,610	38.5	22.4	46.5	12.6	Above-break 3s
5	6,589	28.9	20.7	43.6	6.5	Late-clock 3s

Table 2. Selected GMM clusters ($K=10$). The GNN's latent space recovers recognizable shot archetypes.

Results

Method	AUC	Acc	F1
Static-38 + GMM	0.6551$\pm$.002	62.8%	0.477
XGBoost Vel-65	0.6501 $\pm$.003	62.4%	0.476
XGBoost Static-38	0.6476 $\pm$.003	62.2%	0.476
TemporalGRU (GNN)	0.6451	62.4%	0.445
LR Vel-65 Lasso	0.6377 $\pm$.002	61.2%	0.520
Decision Tree d=7	0.6318 $\pm$.002	61.2%	0.436

Table 3. Top results (87,147 shots; 5-fold CV for baselines; 70/20/10 for GNN).

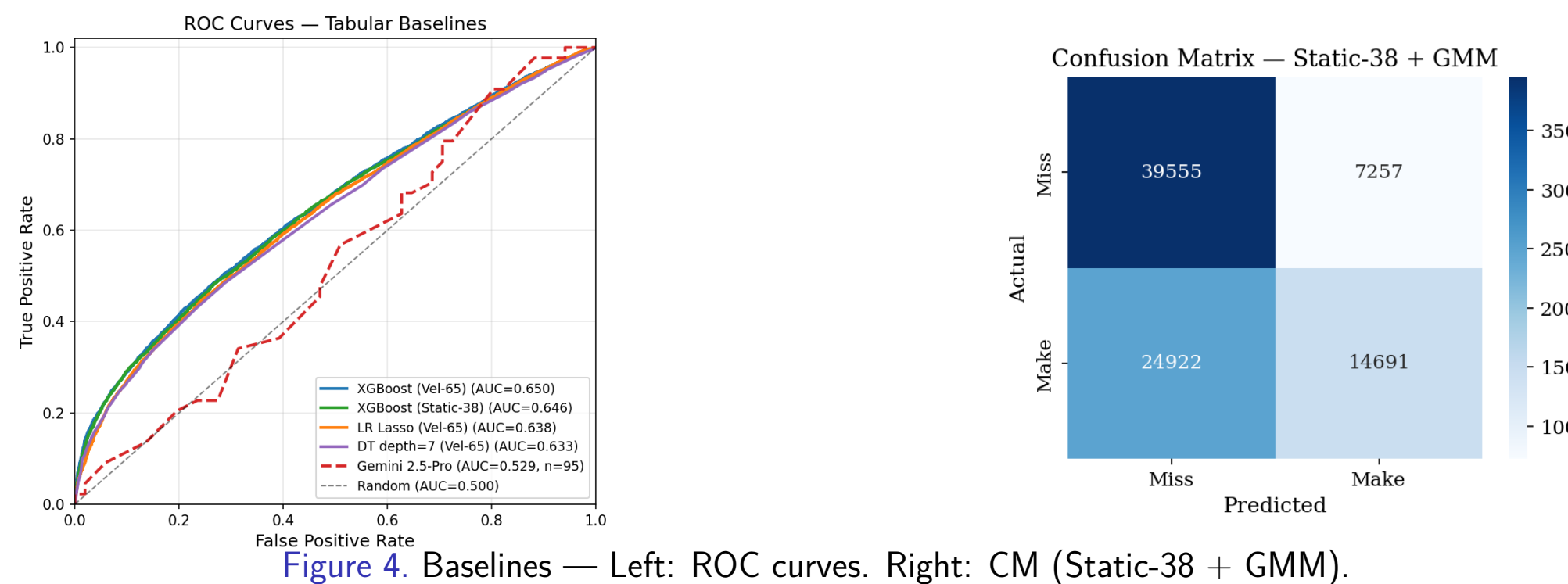


Figure 4. Baselines — Left: ROC curves. Right: CM (Static-38 + GMM).

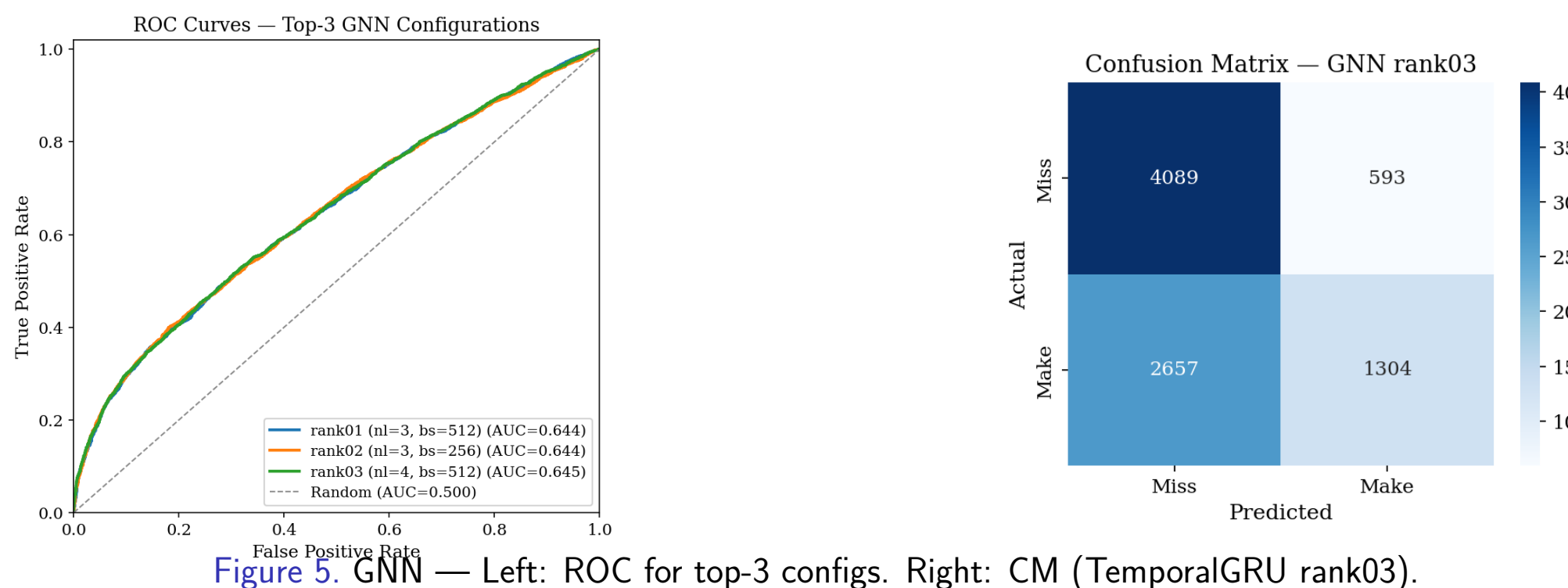


Figure 5. GNN — Left: ROC for top-3 configs. Right: CM (TemporalGRU rank03).

Discussion

The hybrid succeeds because the GNN and XGBoost extract largely **orthogonal** signals: XGBoost relies on **shot distance and shooter FG%**, while the GNN encodes **defender rotations, shot-creation dynamics, and court configuration**. Feature ablation confirms the GNN derives signal from *how the shot is taken* (ΔAUC for player stats = -0.0018 , least important)—nearly the inverse of tabular baselines. Moderate AUC (~ 0.65) reflects basketball's inherent stochasticity (published: 0.59–0.65). HP tuning ($+0.0105$) $>$ architecture; data scaling (1.7K $\rightarrow$ 87K) was the dominant bottleneck.

Future Work

With six more months: (1) replace GRU with a transformer encoder, extending context to 3–5s—our ablation shows a striking interaction effect for richer temporal context; (2) train on multi-season data (2013–16, $\sim 300\text{K}$ shots) for generalization and era-adjusted player evaluation; (3) build a real-time pipeline ingesting live SportVU feeds for coaching decisions and sports-betting calibration.

References

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- Brody, S. et al. “How Attentive are Graph Attention Networks?” *ICLR*, 2022.
- Cho, K. et al. “Learning Phrase Representations using RNN Encoder-Decoder.” *EMNLP*, 2014.